

W13 Exercise

2026.05.21

- The denim dataset concerns the amount of waste in material cutting for a jeans manufacture due to five suppliers.

- 95 observations, 2 variables.

	waste	supplier
1	1.2	1
2	16.4	2
3	12.1	3
4	11.5	4
5	24	5
6	10.1	1
7	-6	2
8	9.7	3
9	10.2	4
10	-3.7	5
...		
95	2.9	2

- An algorithm is used to estimate how much material will be wasted given the dimensions of the material supplied.
- The percentage of waste relative to target was collected weekly for the 5 supplier
- waste=1.2: actual waste is 1.2% higher than the target

- Unbalance design

Supplier	1	2	3	4	5
Num.	22	22	19	19	13

Question

- a) Write down a random-effects model for the denim data. Clearly define all model assumptions and parameters.
- b) Identify the parameters in the model and obtain their REML estimators.
- c) Estimate the effect of each supplier. If only one supplier is to be selected, which supplier would you recommend? Explain your choice.
- d) Construct an approximate 95% confidence interval for the effect of the selected supplier.

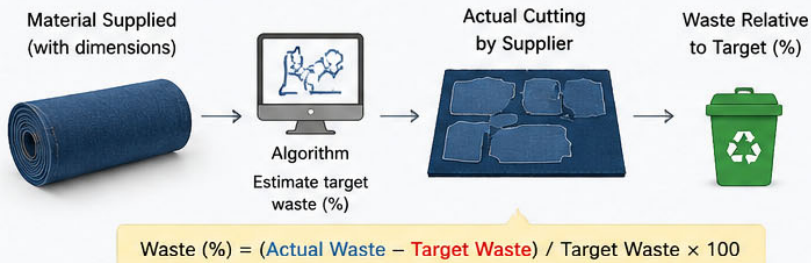


Denim Material Waste Study

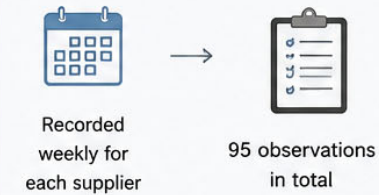
Five suppliers cut denim material for a jeans manufacturer. An algorithm is used to estimate how much material will be wasted given the dimensions of the material supplied. Typically, a supplier wastes more material than the target based on the algorithm although occasionally they waste less. The percentage of waste relative to target was collected weekly for the 5 suppliers. In all, 95 observations were recorded.



1. The Process



2. Data Collection

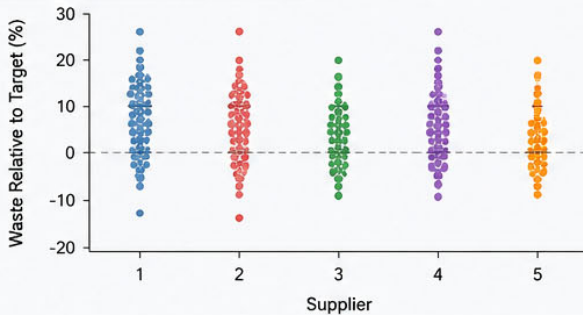


Number of Observations by Supplier

Supplier	1	2	3	4	5
Num.	22	22	19	19	13

Total = 95 observations (Unbalanced Design)

3. What the Data Look Like



Interpretation

- **Positive** value: more waste than target
- **Negative** value: less waste than target
- Suppliers show different average waste levels and variability
- There are both large and small deviations within suppliers

4. Key Questions

- Do the suppliers differ in their average waste relative to the target?
- How large is the variability between suppliers and within suppliers?
- If only one supplier is to be used, which supplier is the best (i.e., has the smallest waste relative to the target)?
- What is the **uncertainty** of the estimated effect of the best supplier?

5. Model Framework (Random-Effects Model)

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

$i = 1, \dots, 5$ suppliers; $j = 1, \dots, n_i$ weeks

$$\begin{cases} \alpha_i \sim N(0, \sigma_\alpha^2) \text{ (supplier effect)} \\ \epsilon_{ij} \sim N(0, \sigma^2) \text{ (within-supplier error)} \\ \alpha_i \text{ and } \epsilon_{ij} \text{ are independent} \end{cases}$$

Parameters

- μ : overall mean waste relative to target
- σ_α^2 : between-supplier variance
- σ^2 : within-supplier variance

Goal: Choose the best supplier and quantify the uncertainty of that choice.